

1. PROJECT & TEAM INFORMATION

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models
- **Team ID:** SWTID-2026-7486
- **Team Leader:** Diwan Oli M

2. CONTEXT DIAGRAM (LEVEL 0 DFD) The Context Diagram illustrates the overarching interactions between the user, the core application, and external cloud services.

- **Source (End-User):** Provides configuration parameters [Prompt, Character, Style, Setting].
- **Central Process (ComicCraft AI System):** Ingests user data, orchestrates API calls, and outputs a [Web Preview] and [Compiled PDF Document].
- **External Service 1 (Google Gemini):** Receives [Text Queries], returns [Structured JSON Layout & Dialogue].
- **External Service 2 (Hugging Face API):** Receives [Image Generation Prompts], returns [Visual Panel Data].

3. SYSTEM BREAKDOWN (LEVEL 1 DFD) This section maps the sequential flow of data across internal application modules.

- **Process 1.0 (User Data Ingestion):**
 - Parses HTTP POST form data.
 - *Data Flow:* Raw HTML Form → Validated Python Dictionary.
- **Process 2.0 (Narrative Structuring):**
 - Transmits the validated prompt to Gemini 1.5 Flash.
 - *Data Flow:* Validated Prompt → 5-Panel JSON Outline.
- **Process 3.0 (Dialogue & Text Generation):**
 - Passes the JSON outline to Gemini 1.5 Pro.
 - *Data Flow:* Outline → Elaborated Comic Script.
- **Process 4.0 (Visual Rendering):**
 - Iteratively queries Stable Diffusion for each panel's scene description.

- *Data Flow*: AI Image Prompts → Downloaded .jpg assets in the static folder.
- **Process 5.0 (Data Aggregation):**
 - Combines the text strings and image file paths into a unified format.
 - *Data Flow*: Text + Image Paths → Comprehensive Layout Dictionary.
- **Process 6.0 (Document Compilation):**
 - Processes the layout dictionary through the FPDF module.
 - *Data Flow*: Layout Dictionary → Final comic_output.pdf.